

Deep Parametric Portfolio Policies

Management Science, 2026

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July 12, 2026

Overview

1. Introduction

2. Theory

3. Design

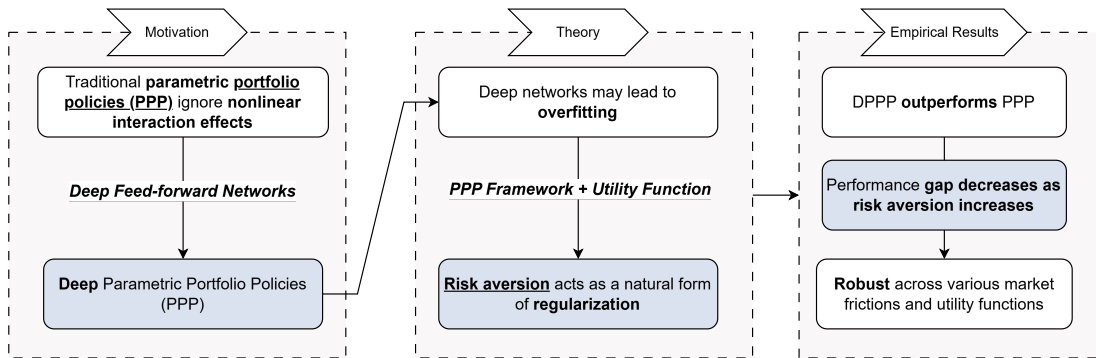
4. Results

5. Replication

6. Ideal

7. Appendix

Summary



Question

1. Do firm characteristics exert **nonlinear, interactive effects** on parametric portfolio weights?
2. Can an investor's **risk aversion coefficient serve as an economic regularization mechanism** to constrain the complexity of complex non-linear models?

Motivation

- Brandt et al. (2009, RFS) propose a **linear parametric portfolio policy (PPP)** that directly maps firm characteristics to portfolio weights

$$w_{i,t} = \bar{w}_{i,t} + \frac{1}{N_t} \theta^\top X_{i,t}, \quad (1)$$

- Avoid mean-variance estimation errors
- **Challenge:** Linear PPP omits **complex nonlinear interactions between characteristics and weights**
 - Nonlinear interactions between features and expected returns (Gu et al., 2020, RFS)
- Solution: Use **deep feedforward networks** to model these nonlinear relationships
- **How to control deep network complexity?** The **risk aversion** coefficient can serve as economic regularization

Marginal Contributions

- **Model Regularization**

- **Prior work:** **Statistical regularization** (e.g., L1 and L2 penalties) can reduce model complexity (Gu et al., 2020, RFS; Bianchi et al., 2021, RFS; Chen et al., 2023, MS)
- **This paper:** Theoretical and empirical demonstration that investor **risk aversion serves as a form of economic regularization**, exerting a similar effect

- **Parametric Portfolio Policy (PPP)**

- **Prior work:** **Under a linear framework**, discuss the impact of leverage constraints (Ammann et al., 2016, JBF), transaction costs (DeMigue et al., 2020, RFS), and estimation risk (Lamoureux et al., 2024, RAPS) on weight estimation
- **This paper:** Expand the traditional linear PPP into a **non-linear Deep PPP** model

Risk Aversion and Weight Convergence

- **Portfolio Return Definition.** Based on the parametric portfolio policy of Brandt et al. (2009), the **portfolio return defined** as:

$$r_{p,t+1}(\theta) = \bar{w}_t^T r_{t+1} + \theta^T X_t^T r_{t+1} / N_t = r_{\bar{w},t+1} + \theta^T r_{c,t+1} \quad (2)$$

- **Utility Approximation.** A 2nd-order Taylor expansion of **CRRA utility** defined as:

$$\mathbb{E}[u(r_{p,t+1}(\theta))] \approx \mathbb{E}[r_{p,t+1}(\theta)] - \frac{\gamma}{2} \mathbb{E}[r_{p,t+1}(\theta)^2] \quad (3)$$

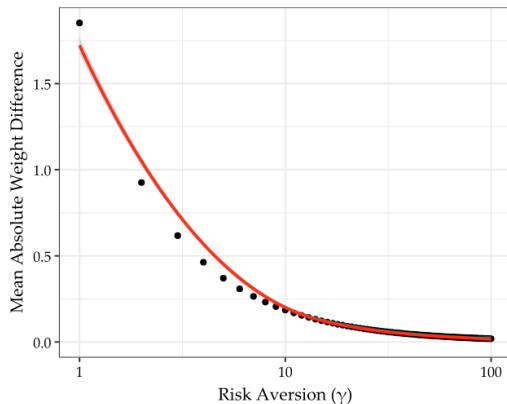
- **Model Convergence.** By substituting Equation (1) into Equation (2) and taking the first-order condition with respect to θ :

$$\lim_{\gamma \rightarrow \infty} \|w_{PPP}(\gamma) - w_{DPPP}(\gamma)\| = \underbrace{\lim_{\gamma \rightarrow \infty} \frac{1}{\gamma} \|\Delta_{RP}\|}_{=0} + \lim_{\gamma \rightarrow \infty} \|\Delta_{RM}\| \quad (4)$$

Economic Intuition: Risk-averse investors naturally constrain model complexity to mitigate the excess volatility induced by estimation risk

Risk Aversion and Weight Convergence

- **Mean absolute difference in portfolio weights** between PPP (linear) and DPPP (nonlinear) models across risk aversion levels



Loss Function

- Minimize the constant relative risk aversion (CRRA) utility with respect to θ :

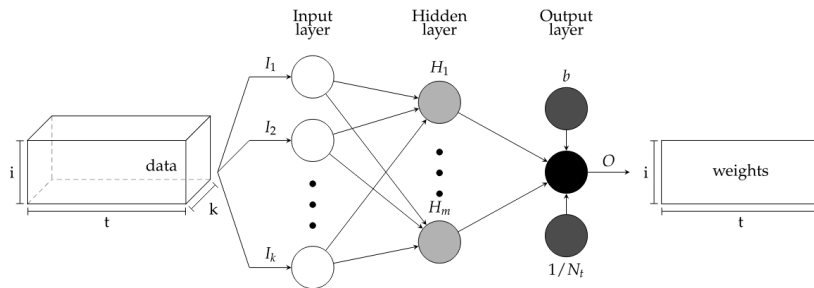
$$\mathcal{L}(\theta) = -\frac{1}{T} \sum_{t=1}^T \left(\frac{(1 + r_{p,t+1}(\theta))^{1-\gamma}}{1-\gamma} \right) \quad (5)$$

- Minimizing Equation (5) is equivalent to maximizing CRRA utility

Feed-forward Neural Network Structure

- 3–5 hidden layers (selected via validation set)
- Maps each feature vector $X_{i,t}$ to an active weight $f(X_{i,t}; \theta)$

$$w_{i,t} = \bar{w}_{i,t} + \frac{1}{N_t} f(X_{i,t}; \theta) \quad (6)$$



Data and Estimate Strategy

- 157 predictors from the dataset of Chen and Zimmermann (2022)
- January 1971 to December 2020
- Expanding training window (initial 20 years, updated monthly) with a fixed 5-year validation and 12-month out-of-sample prediction

Performance Results for CRRA Investors

1. The DPPP **consistently outperforms** the PPP
2. The **performance differential decline with increasing risk aversion**
3. The **volatility** of DPPP **declines with increasing risk aversion**

	$\gamma = 2$			$\gamma = 5$			$\gamma = 10$		
	PPP	DPPP	Δ	PPP	DPPP	Δ	PPP	DPPP	Δ
CER	0.0195	0.0297	0.0102	0.0163	0.0232	0.0069	0.0109	0.0152	0.0043
$p\text{-value}(\text{CER}_{DPPP} - \text{CER}_{PPP})$		0.0003			0.0002			0.0338	
$\sum w_i /N_t * 100$	0.1696	0.1907	0.0211	0.1770	0.1938	0.0168	0.1769	0.1933	0.0164
Mean	0.0220	0.0341	0.0121	0.0214	0.0305	0.0091	0.0201	0.0281	0.0080
StdDev	0.0492	0.0710	0.0218	0.0435	0.0550	0.0115	0.0401	0.0475	0.0074
Max DD	0.6302	0.4979	-0.1323	0.4467	0.5601	0.1134	0.3953	0.4662	0.0709
SR	1.5465	1.6607	0.1142	1.7007	1.9230	0.2223	1.7404	2.0446	0.3042
$p\text{-value}(\text{SR}_{DPPP} - \text{SR}_{PPP})$		0.3709			0.0985			0.0445	

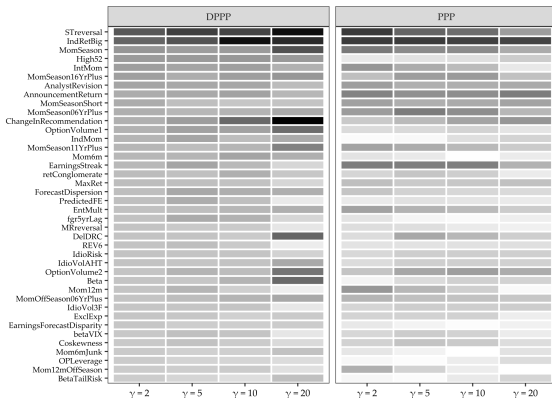
Market Frictions

- **Market frictions** naturally act as a form of **"economic regularization,"** thereby narrowing the differences between models

	$\gamma = 2$			$\gamma = 5$			$\gamma = 10$		
	PPP	DPPP	Δ	PPP	DPPP	Δ	PPP	DPPP	Δ
Transaction costs									
CER	0.0155	0.0194	0.0039	0.0129	0.0157	0.0028	0.0077	0.0087	0.0010
$p\text{-value}(\text{CER}_{DPPP} - \text{CER}_{PPP})$		0.0118			0.0620			0.3195	
Mean	0.0179	0.0225	0.0046	0.0178	0.0221	0.0043	0.0170	0.0182	0.0012
StdDev	0.0482	0.0551	0.0069	0.0427	0.0498	0.0071	0.0397	0.0412	0.0015
Max 1M loss	0.2228	0.2280	0.0052	0.1812	0.2015	0.0203	0.1559	0.1546	-0.0013
SR	1.2851	1.4123	0.1272	1.4453	1.5370	0.0917	1.4823	1.5296	0.0473
$p\text{-value}(\text{SR}_{DPPP} - \text{SR}_{PPP})$		0.2090			0.2962			0.3852	
Long-only									
CER	0.0118	0.0164	0.0046	0.0076	0.0107	0.0031	0.0011	0.0020	0.0009
$p\text{-value}(\text{CER}_{DPPP} - \text{CER}_{PPP})$		0.0001			0.0143			0.3308	
Mean	0.0150	0.0215	0.0065	0.0147	0.0216	0.0069	0.0137	0.0174	0.0037
StdDev	0.0566	0.0713	0.0147	0.0510	0.0647	0.0137	0.0459	0.0490	0.0031
Max 1M loss	0.2483	0.2603	0.0120	0.2171	0.2667	0.0496	0.1968	0.2260	0.0292
SR	0.9213	1.0418	0.1205	0.9996	1.1580	0.1584	1.0342	1.2262	0.1920
$p\text{-value}(\text{SR}_{DPPP} - \text{SR}_{PPP})$		0.0119			0.0077			0.0006	

Variable Importance

- **Past return-based characteristics** dominate the importance rankings
- DPPP relies on a **narrower set of features (feature selection)**, an effect **amplified by higher risk aversion**



Performance Results for CRRA Investors (US)

	Original						Replication (US)					
	$\gamma = 2$			$\gamma = 10$			$\gamma = 2$			$\gamma = 10$		
	PPP	DPPP	Δ	PPP	DPPP	Δ	PPP	DPPP	Δ	PPP	DPPP	Δ
CER	0.0195	0.0297	0.0102	0.0109	0.0152	0.0043	0.0171	0.0254	0.0083	0.0101	0.0162	0.0061
$p\text{-value}(\text{CER}_{DPPP} - \text{CER}_{PPP})$		0.0003			0.0338			0.0003			0.0008	
$\sum w_i / N_t * 100$	0.1696	0.1907	0.0211	0.1769	0.1933	0.0164	0.1694	0.1922	0.0228	0.1735	0.1971	0.0236
Mean	0.0220	0.0341	0.0121	0.0201	0.0281	0.0080	0.0193	0.0289	0.0096	0.0192	0.0272	0.0080
StdDev	0.0492	0.0710	0.0218	0.0401	0.0475	0.0074	0.0460	0.0597	0.0137	0.0398	0.0435	0.0037
Max DD	0.6302	0.4979	-0.1323	0.3953	0.4662	0.0709	0.4714	0.6059	0.1345	0.4320	0.4027	-0.0293
SR	1.5465	1.6607	0.1142	1.7404	2.0446	0.3042	1.4527	1.6780	0.2253	1.6685	2.1668	0.4983
$p\text{-value}(\text{SR}_{DPPP} - \text{SR}_{PPP})$		0.3709			0.0445			0.0422			0.0012	

Market Frictions (US)

	Original						Replication (US)					
	$\gamma = 2$			$\gamma = 10$			$\gamma = 2$			$\gamma = 10$		
	PPP	DPPP	Δ	PPP	DPPP	Δ	PPP	DPPP	Δ	PPP	DPPP	Δ
Transaction costs												
CER	0.0155	0.0194	0.0039	0.0077	0.0087	0.0010	0.0115	0.0154	0.0039	0.0042	0.0078	0.0036
p -value($CER_{DPPP} - CER_{PPP}$)		0.0118			0.3195			0.0036			0.0030	
Mean	0.0179	0.0225	0.0046	0.0170	0.0182	0.0012	0.0140	0.0186	0.0046	0.0139	0.0168	0.0029
StdDev	0.0482	0.0551	0.0069	0.0397	0.0412	0.0015	0.0484	0.0559	0.0075	0.0402	0.0401	-0.0001
Max 1M loss	0.2228	0.2280	0.0052	0.1559	0.1546	-0.0013	0.2269	0.2315	0.0046	0.1740	0.1494	-0.0246
SR	1.2851	1.4123	0.1272	1.4823	1.5296	0.0473	1.0000	1.1489	0.1489	1.1934	1.4479	0.2545
p -value($SR_{DPPP} - SR_{PPP}$)		0.2090			0.3852			0.0486			0.0069	
Long-only												
CER	0.0118	0.0164	0.0046	0.0011	0.0020	0.0009	0.0112	0.0135	0.0023	0.0002	0.0002	0.0000
p -value($CER_{DPPP} - CER_{PPP}$)		0.0001			0.3308			0.0058			0.4306	
Mean	0.0150	0.0215	0.0065	0.0137	0.0174	0.0037	0.0145	0.0177	0.0032	0.0132	0.0156	0.0024
StdDev	0.0566	0.0713	0.0147	0.0459	0.0490	0.0031	0.0565	0.0646	0.0081	0.0461	0.0499	0.0038
Max 1M loss	0.2483	0.2603	0.0120	0.1968	0.2260	0.0292	0.2402	0.2577	0.0175	0.1983	0.2180	0.0197
SR	0.9213	1.0418	0.1205	1.0342	1.2262	0.1920	0.8870	0.9497	0.0627	0.9964	1.0871	0.0907
p -value($SR_{DPPP} - SR_{PPP}$)		0.0119			0.0006			0.0478			0.0101	

Variable Importance (US)

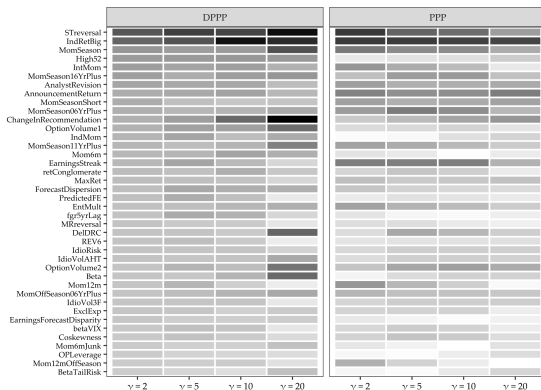


Figure: Original

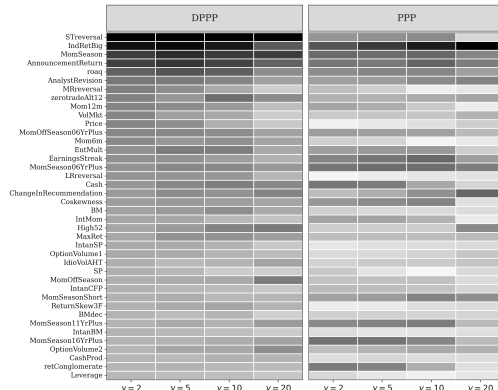


Figure: Replication (US)

Performance Results for CRRA Investors (CHN)

	Original						Replication (CHN)					
	$\gamma = 2$			$\gamma = 10$			$\gamma = 2$			$\gamma = 10$		
	PPP	DPPP	Δ	PPP	DPPP	Δ	PPP	DPPP	Δ	PPP	DPPP	Δ
CER	0.0195	0.0297	0.0102	0.0109	0.0152	0.0043	0.0203	0.0428	0.0225	-0.0152	0.0041	0.0193
$p\text{-value}(\text{CER}_{DPPP} - \text{CER}_{PPP})$		0.0003			0.0338			0.0001			0.0001	
$\sum w_i /N_t * 100$	0.1696	0.1907	0.0211	0.1769	0.1933	0.0164	0.0683	0.0923	0.0240	0.0645	0.0796	0.0151
Mean	0.0220	0.0341	0.0121	0.0201	0.0281	0.0080	0.0282	0.0515	0.0233	0.0259	0.0427	0.0168
StdDev	0.0492	0.0710	0.0218	0.0401	0.0475	0.0074	0.0908	0.0973	0.0065	0.0867	0.0891	0.0024
Max DD	0.6302	0.4979	-0.1323	0.3953	0.4662	0.0709	0.9821	0.7946	-0.1875	1.0058	0.8192	-0.1866
SR	1.5465	1.6607	0.1142	1.7404	2.0446	0.3042	1.0765	1.8340	0.7575	1.0341	1.6603	0.6262
$p\text{-value}(\text{SR}_{DPPP} - \text{SR}_{PPP})$		0.3709			0.0445			0.0001			0.0001	

Market Frictions (CHN)

	Original						Replication (CHN)					
	$\gamma = 2$			$\gamma = 10$			$\gamma = 2$			$\gamma = 10$		
	PPP	DPPP	Δ	PPP	DPPP	Δ	PPP	DPPP	Δ	PPP	DPPP	Δ
Transaction costs												
CER	0.0155	0.0194	0.0039	0.0077	0.0087	0.0010	0.0148	0.0333	0.0185	-0.0193	-0.0040	0.0153
p -value($CER_{DPPP} - CER_{PPP}$)		0.0118			0.3195			0.0001			0.0001	
Mean	0.0179	0.0225	0.0046	0.0170	0.0182	0.0012	0.0231	0.0419	0.0188	0.0219	0.0336	0.0117
StdDev	0.0482	0.0551	0.0069	0.0397	0.0412	0.0015	0.0924	0.0961	0.0037	0.0867	0.0874	0.0007
Max 1M loss	0.2228	0.2280	0.0052	0.1559	0.1546	-0.0013	0.2409	0.2023	-0.0386	0.2435	0.2072	-0.0363
SR	1.2851	1.4123	0.1272	1.4823	1.5296	0.0473	0.8649	1.5102	0.6453	0.8751	1.3330	0.4579
p -value($SR_{DPPP} - SR_{PPP}$)		0.2090			0.3852			0.0001			0.0001	
Long-only												
CER	0.0118	0.0164	0.0046	0.0011	0.0020	0.0009	0.0135	0.0184	0.0049	-0.0257	-0.0194	0.0063
p -value($CER_{DPPP} - CER_{PPP}$)		0.0001			0.3308			0.0001			0.0001	
Mean	0.0150	0.0215	0.0065	0.0137	0.0174	0.0037	0.0221	0.0271	0.0050	0.0196	0.0236	0.0040
StdDev	0.0566	0.0713	0.0147	0.0459	0.0490	0.0031	0.0935	0.0942	0.0007	0.0894	0.0892	-0.0002
Max 1M loss	0.2483	0.2603	0.0120	0.1968	0.2260	0.0292	0.2688	0.2553	-0.0135	0.2676	0.2523	-0.0153
SR	0.9213	1.0418	0.1205	1.0342	1.2262	0.1920	0.8171	0.9970	0.1799	0.7595	0.9156	0.1561
p -value($SR_{DPPP} - SR_{PPP}$)		0.0119			0.0006			0.0001			0.0001	

Variable Importance (CHN)

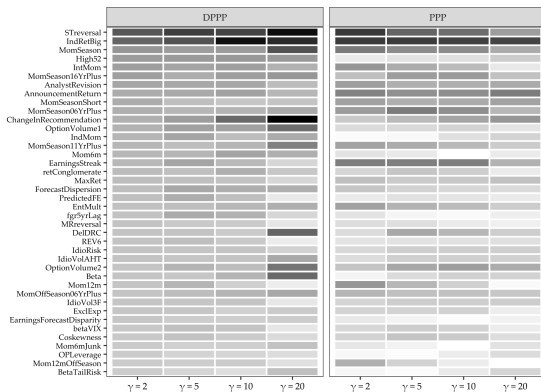


Figure: Original

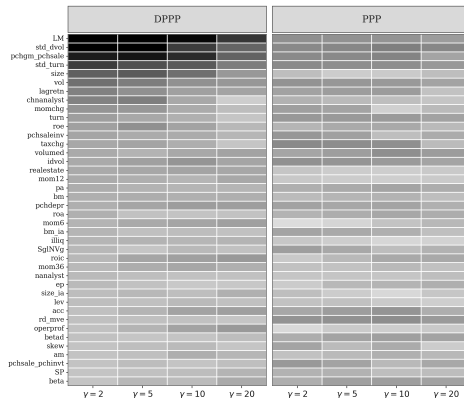


Figure: Replication (CHN)

Extension

- Accommodate **diverse investor preferences**
 - Provide customized portfolio allocations by adapting the objective function to reflect specific investor preferences (e.g., transaction costs, extreme returns, or subjective discount rates)
- Maximize the **expected lifetime utility** of the investor

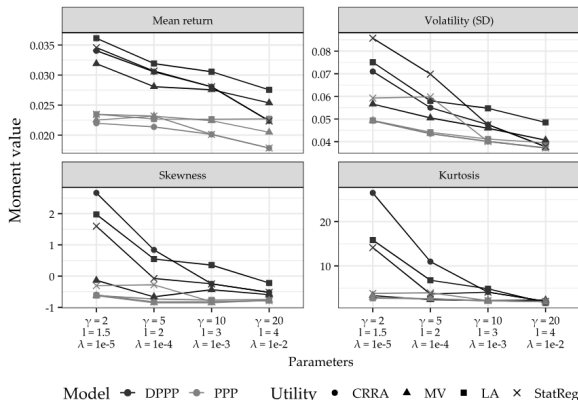
Alternative Investor Utility Functions

- DPPP's nonlinear features provide a distinct advantage in capturing complex return distributions, which is crucial for investors sensitive to higher-order moments

Mean-variance preference	$\gamma = 2$			$\gamma = 5$			$\gamma = 10$		
	PPP	DPPP	Δ	PPP	DPPP	Δ	PPP	DPPP	Δ
CER	0.0201	0.0287	0.0086	0.0184	0.0217	0.0033	0.0143	0.0170	0.0027
$p\text{-value}(\text{CER}_{DPPP} - \text{CER}_{PPP})$		0.0001			0.0292			0.0291	
Mean	0.0225	0.0319	0.0094	0.0232	0.0281	0.0049	0.0224	0.0276	0.0052
StdDev	0.0492	0.0566	0.0074	0.0435	0.0505	0.0070	0.0402	0.0459	0.0057
Skew	-0.6239	-0.1348	0.4891	-0.8530	-0.6631	0.1899	-0.8516	-0.4331	0.4185
SR	1.5843	1.9506	0.3663	1.8438	1.9259	0.0821	1.9317	2.0786	0.1469
$p\text{-value}(\text{SR}_{DPPP} - \text{SR}_{PPP})$		0.0019			0.2768			0.1185	
Loss aversion preference	$l = 1.5$			$l = 2$			$l = 3$		
	PPP	DPPP	Δ	PPP	DPPP	Δ	PPP	DPPP	Δ
CER	0.0188	0.0311	0.0123	0.0147	0.0235	0.0088	0.0082	0.0137	0.0055
$p\text{-value}(\text{CER}_{DPPP} - \text{CER}_{PPP})$		0.0002			0.0015			0.0247	
Mean	0.0235	0.0361	0.0126	0.0227	0.0319	0.0092	0.0226	0.0306	0.0080
StdDev	0.0494	0.0751	0.0257	0.0442	0.0580	0.0138	0.0412	0.0548	0.0136
Skew	-0.6194	1.9765	2.5959	-0.7339	0.5481	1.2820	-0.7651	0.3536	1.1187
SR	1.6475	1.6666	0.0191	1.7793	1.9049	0.1256	1.9052	1.9331	0.0279
$p\text{-value}(\text{SR}_{DPPP} - \text{SR}_{PPP})$		0.4931			0.2424			0.4498	

Portfolio Moments Across Utility Functions and Models

- Increasing risk or loss aversion systematically reduces portfolio volatility and higher-order moment exposures



Comparison with Benchmark Models

- DPPP strategies deliver highest outof-sample Sharpe ratios

	$\gamma = 2$	$\gamma = 5$	$\gamma = 10$	$\gamma = 20$	SR_{PPP}	SR_{DPPP}	ML	Factor	EW	VW
Mean	0.0341	0.0305	0.0281	0.0224	0.0202	0.0192	0.0193	0.0115	0.0110	0.0105
StdDev	0.0710	0.0550	0.0475	0.0378	0.0354	0.0354	0.0667	0.0416	0.0587	0.0552
Skew	2.6646	0.8411	-0.2470	-0.5201	-0.7604	-0.6342	-0.0544	-1.0598	-0.3716	-0.5039
Kurt	26.4755	10.9695	4.0705	1.9954	1.8370	1.8973	7.2504	3.2232	3.6591	3.3455
Max DD	0.4979	0.5601	0.4662	0.3027	0.2607	0.4356	1.0025	0.7836	0.9026	0.8585
Max 1M loss	0.2264	0.1789	0.1838	0.1446	0.1304	0.1336	0.3531	0.1924	0.2556	0.2398
CVaR (95%)	0.1107	0.0978	0.0882	0.0713	0.0732	0.0693	0.1466	0.1037	0.1351	0.1286
SR	1.6607	1.9230	2.0446	2.0491	1.9783	1.8815	1.0042	0.9590	0.6461	0.6609
<i>FF5 + Mom α</i>	0.0232	0.0205	0.0182	0.0130	0.0110	0.0102	0.0048	0.0010	-0.0002	-0.0003
<i>StdErr(α)</i>	0.0029	0.0024	0.0020	0.0016	0.0015	0.0016	0.0019	0.0009	0.0007	0.0006

Certainty Equivalent Return (CER)

CRRA utility is specified as:

$$U(1 + R_p) = \frac{(1 + R_p)^{1-\gamma}}{1 - \gamma} \quad (7)$$

According to the theoretical definition of CER:

$$U(1 + CER) = \mathbb{E}[U(1 + R_p)] \quad (8)$$

In empirical estimation, the out-of-sample CER is derived as:

$$CER = \left(\frac{1}{T} \sum_{t=1}^T (1 + R_{p,t})^{1-\gamma} \right)^{\frac{1}{1-\gamma}} - 1 \quad (9)$$

Derivation of Model Convergence

- **Portfolio Return Definition.** Following the parametric portfolio policy of Brandt et al. (2009), the return is a combination of an equally-weighted portfolio and an active characteristic-based tilt:

$$r_{p,t+1}(\theta) = \bar{w}_t^T r_{t+1} + \theta^T X_t^T r_{t+1} / N_t = r_{\bar{w},t+1} + \theta^T r_{c,t+1} \quad (10)$$

- **CRRA Utility Expansion.** A 2nd-order Taylor expansion of CRRA utility penalizes portfolio variance via risk aversion γ , simplifying to a Mean-Variance framework:

$$\mathbb{E}[u(r_{p,t+1}(\theta))] \approx \mathbb{E}[r_{p,t+1}(\theta)] - \frac{\gamma}{2} \mathbb{E}[r_{p,t+1}(\theta)^2] \quad (11)$$

Derivation of Model Convergence

- **First-Order Condition.** Substituting Eq. (1) into Eq. (2) and optimizing with respect to θ yields the active deviations. Here, S_t is defined as a nonlinear mapping of the original features X_t :

$$\frac{1}{N_t} \theta^{*T} S_t = \underbrace{\frac{1}{N_t} \frac{1}{\gamma} \hat{\Sigma}_c^{-1} \hat{\mu}_c^T S_t}_{\text{Risk premium term}} - \underbrace{\frac{1}{N_t} \hat{\Sigma}_c^{-1} \hat{\sigma}_{bc}^T S_t}_{\text{Risk minimization term}} \quad (12)$$

- **Extreme Risk Aversion.** As risk aversion increases ($\gamma \rightarrow \infty$), the premium-seeking behavior vanishes completely:

$$\lim_{\gamma \rightarrow \infty} \frac{1}{N_t} \theta^T S_t = - \frac{1}{N_t} \hat{\Sigma}_c^{-1} \hat{\sigma}_{bc}^T S_t \quad (13)$$

Derivation of Model Convergence

- **Structural Convergence.** Evaluating the absolute weight difference between the linear (PPP) and nonlinear (DPPP) models:

$$\lim_{\gamma \rightarrow \infty} \|w_{PPP}(\gamma) - w_{DPPP}(\gamma)\| = \underbrace{\lim_{\gamma \rightarrow \infty} \frac{1}{\gamma} \|\Delta_{RP}\|}_{=0} + \lim_{\gamma \rightarrow \infty} \|\Delta_{RM}\| \quad (14)$$

- Where the structural differences in the risk premium (Δ_{RP}) and risk minimization (Δ_{RM}) components are explicitly defined as:

$$\Delta_{RP} = \hat{\Sigma}_X^{-1} \hat{\mu}_X^T X_t - \hat{\Sigma}_S^{-1} \hat{\mu}_S^T S_t$$

$$\Delta_{RM} = \hat{\Sigma}_X^{-1} \hat{\sigma}_{bX}^T X_t - \hat{\Sigma}_S^{-1} \hat{\sigma}_{bS}^T S_t$$

- The difference strictly converges to $\|\Delta_{RM}\|$, a constant hedging difference that is structurally independent of γ .

Parametric Portfolio Policies: Mean-Variance Solution

Defining the objective function:

$$U = E[R_p] - \frac{\gamma}{2} \text{Var}(R_p) \quad (15)$$

Substituting the portfolio return $R_p = R_{bench} + \theta^\top R_{port}$:

$$U = (\mu_{bench} + \theta^\top \mu_{port}) - \frac{\gamma}{2} (\sigma_{bench}^2 + \theta^\top \Sigma_{port} \theta + 2\theta^\top \Sigma_{bench,port}) \quad (16)$$

Setting $\frac{\partial U}{\partial \theta} = 0$ provides the analytical solution for θ :

$$\theta = \frac{1}{\gamma} \Sigma_{port}^{-1} \mu_{port} - \Sigma_{port}^{-1} \Sigma_{bench,port} \quad (17)$$

- This approach maximizes computational efficiency and enables the exact calculation of effective degrees of freedom (EDF).